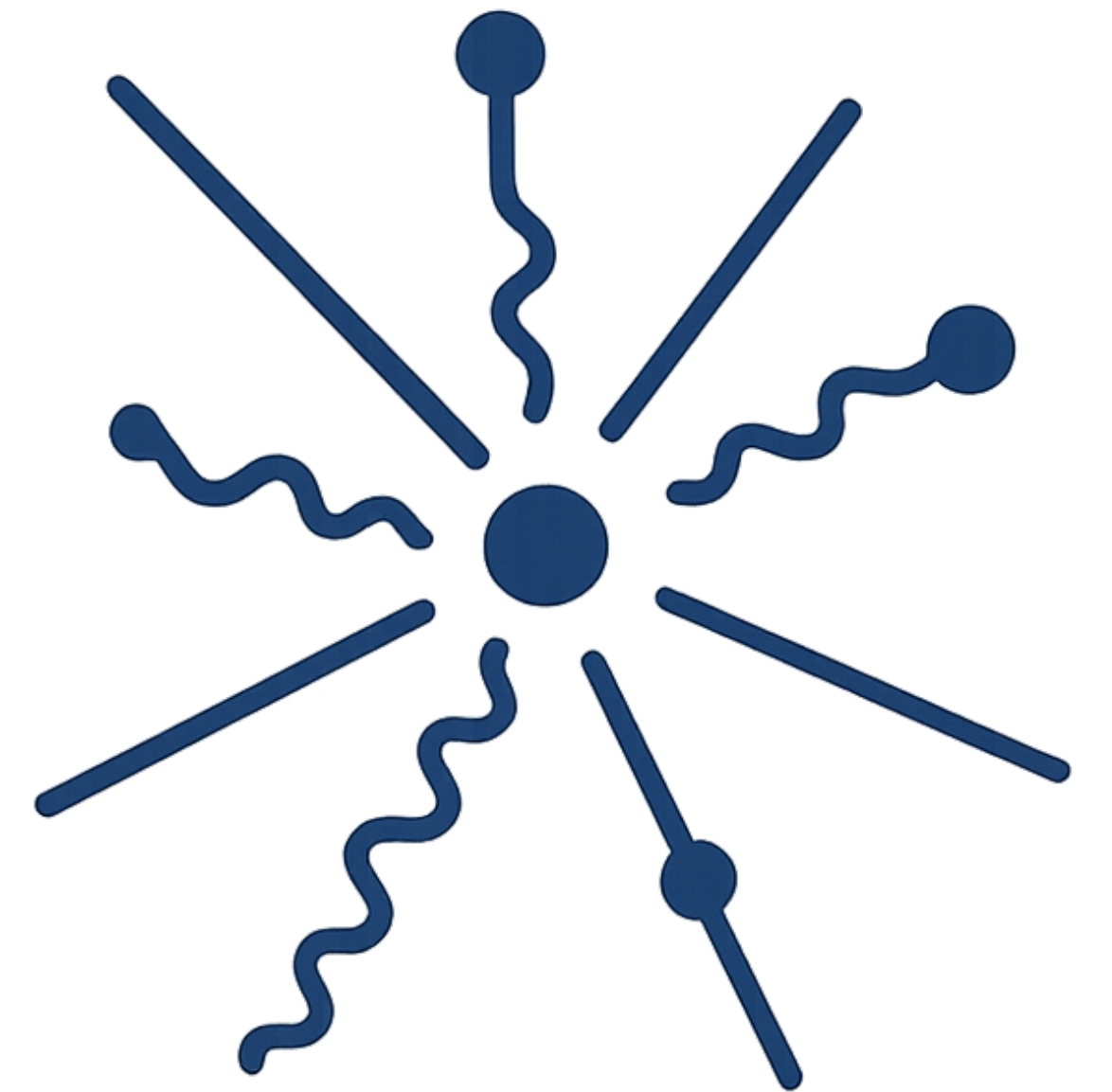
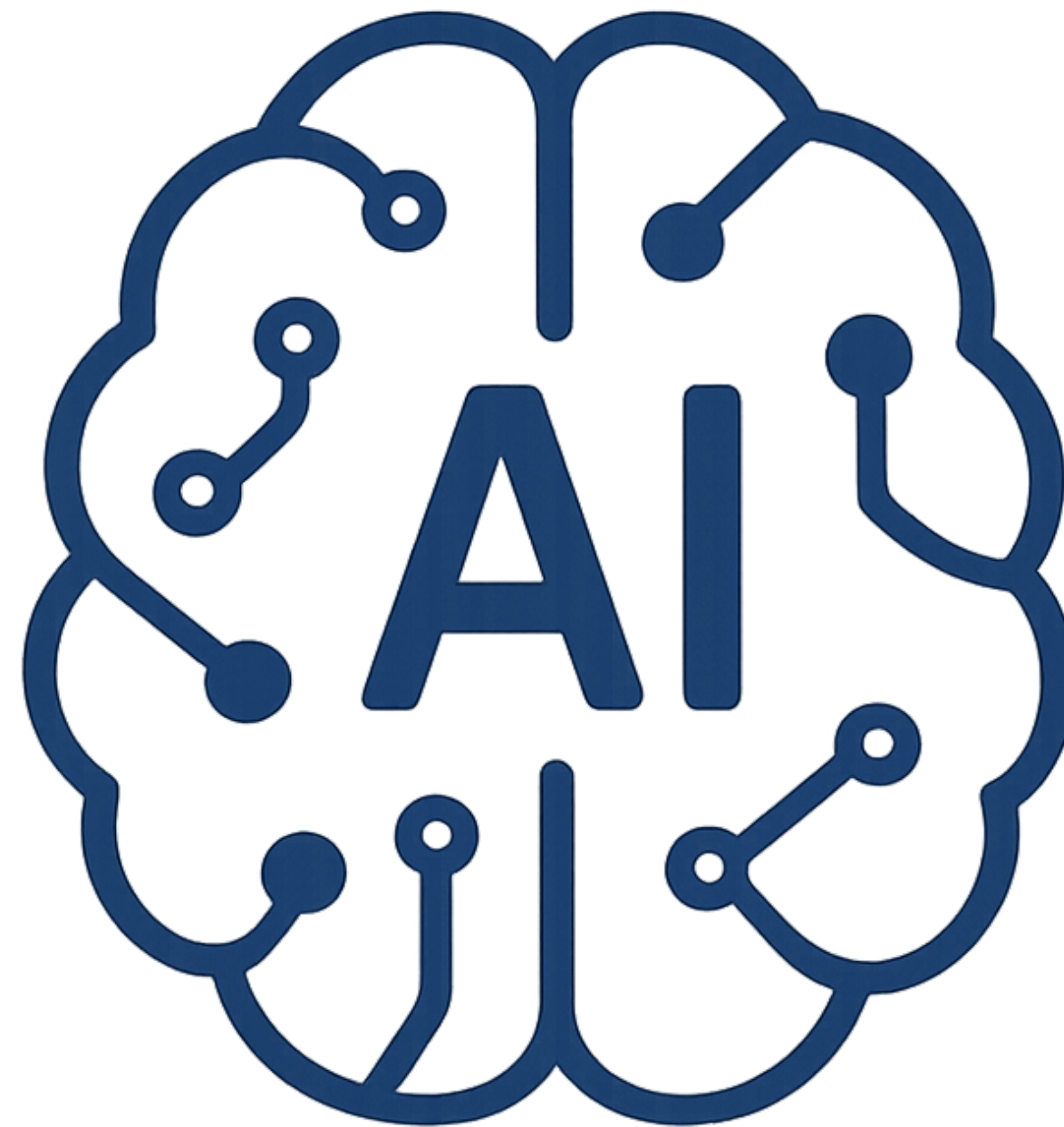
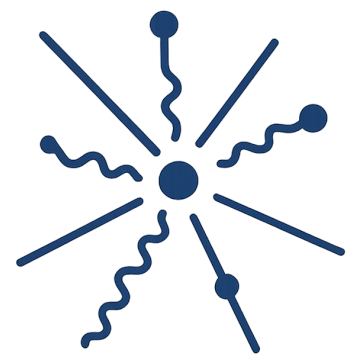
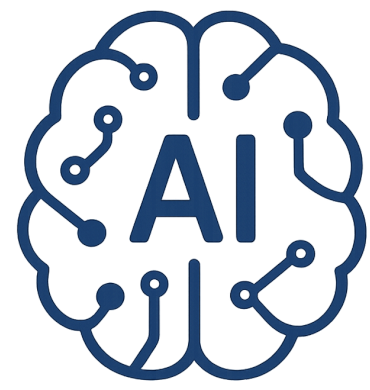


Introduction to

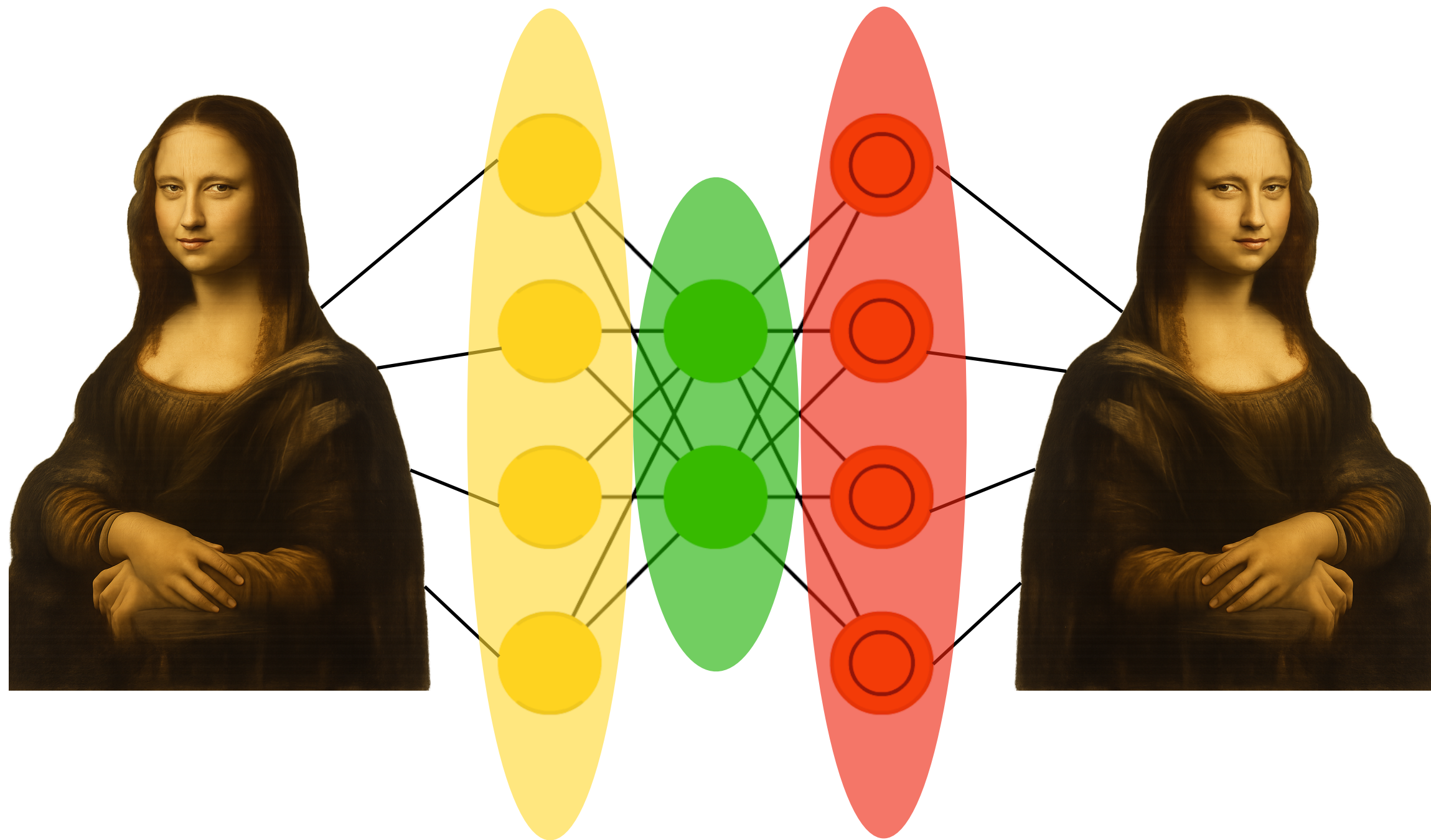
AI-Driven HEP

S. A. Fard - School of Physics (IPM)





AI-Driven HEP 20



Session 20

Latent Representation

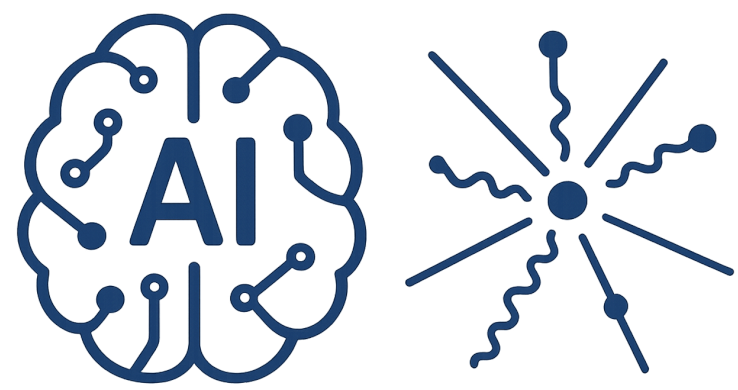


Reference

- [1] Bishop, Christopher M., and Hugh Bishop. Deep learning: Foundations and concepts. Springer Nature, 2023.
- [2] AN IMAGE IS WORTH 16X16 WORDS: TRANSFORMERS FOR IMAGE RECOGNITION AT SCALE, <https://arxiv.org/pdf/2010.11929v2>
- [3] Masked Autoencoders Are Scalable Vision Learners, <https://arxiv.org/abs/2111.06377v3>

<https://physics.ipm.ac.ir/~ansarifard/>



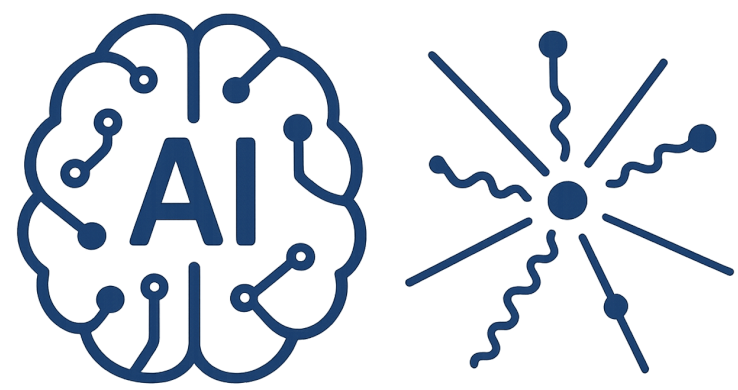


AI-Driven HEP 20

Latent Space

Try to uncover the masked area





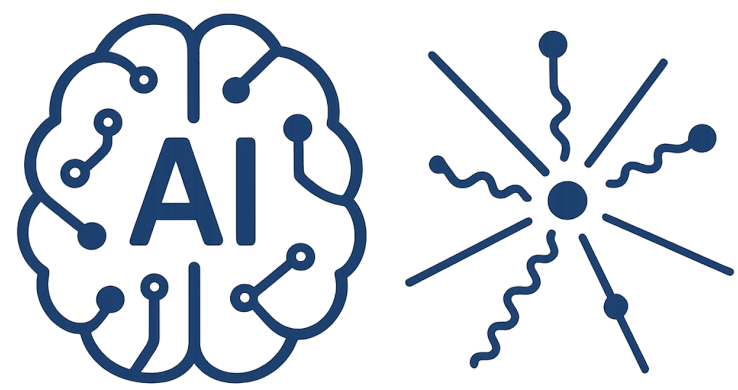
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Latent Space

Try to uncover the masked area

How it works?





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Latent Space

Try to uncover the masked area

How it works?

internal encoding of data

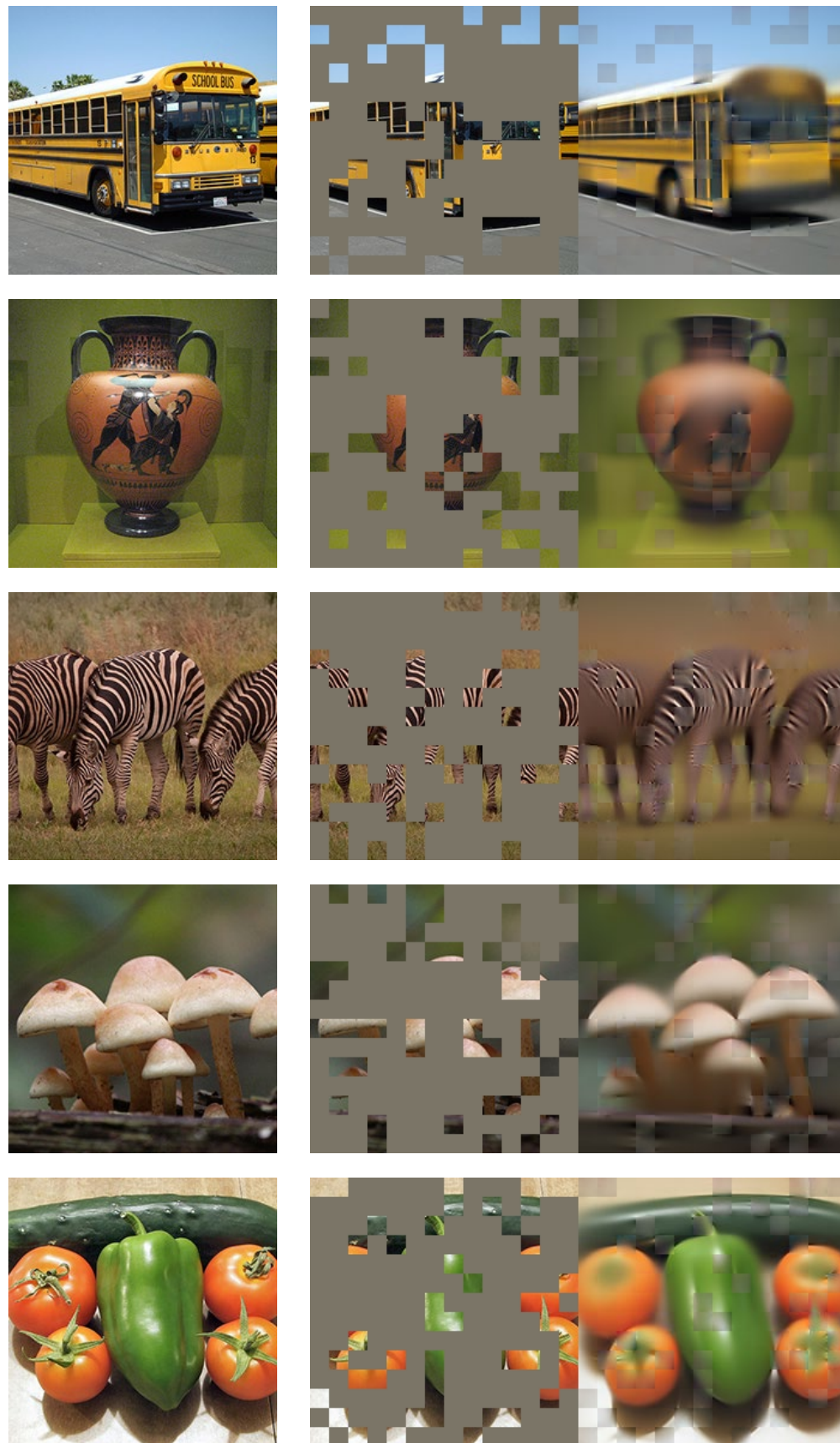


Generative Tasks

Modeling how data is created

$$p(x \mid \text{context})?$$

Learn the underlying data distribution to produce new sample



original

mask 75%

Generative Tasks

Modeling how data is created

$$p(x \mid \text{context})?$$

Learn the underlying data distribution to produce new sample

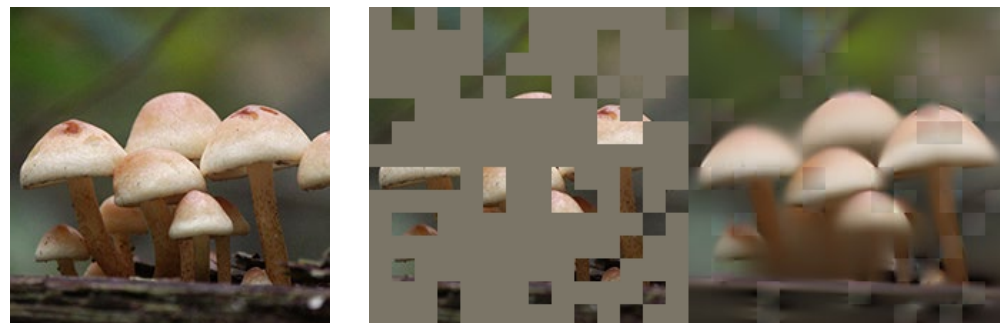
In General : Central to foundation models that transfer knowledge across tasks and domains



original

mask 75%

Generative Tasks



original

mask 75%

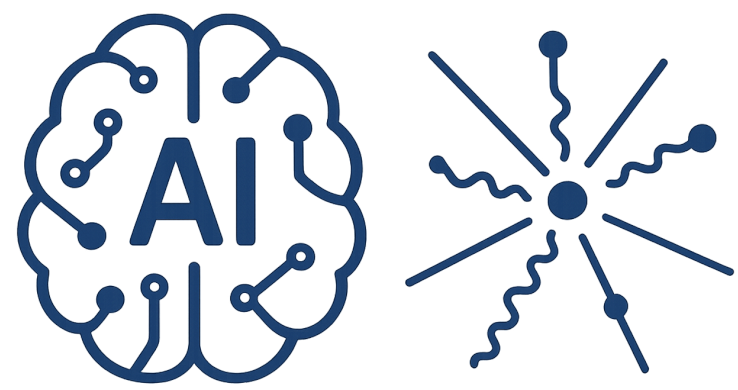
Modeling how data is created

$$p(x \mid \text{context})?$$

Learn the underlying data distribution to produce new sample

In General : Central to foundation models that transfer knowledge across tasks and domains

In physics : Experiment (Observation) Simulation, New Physics searches

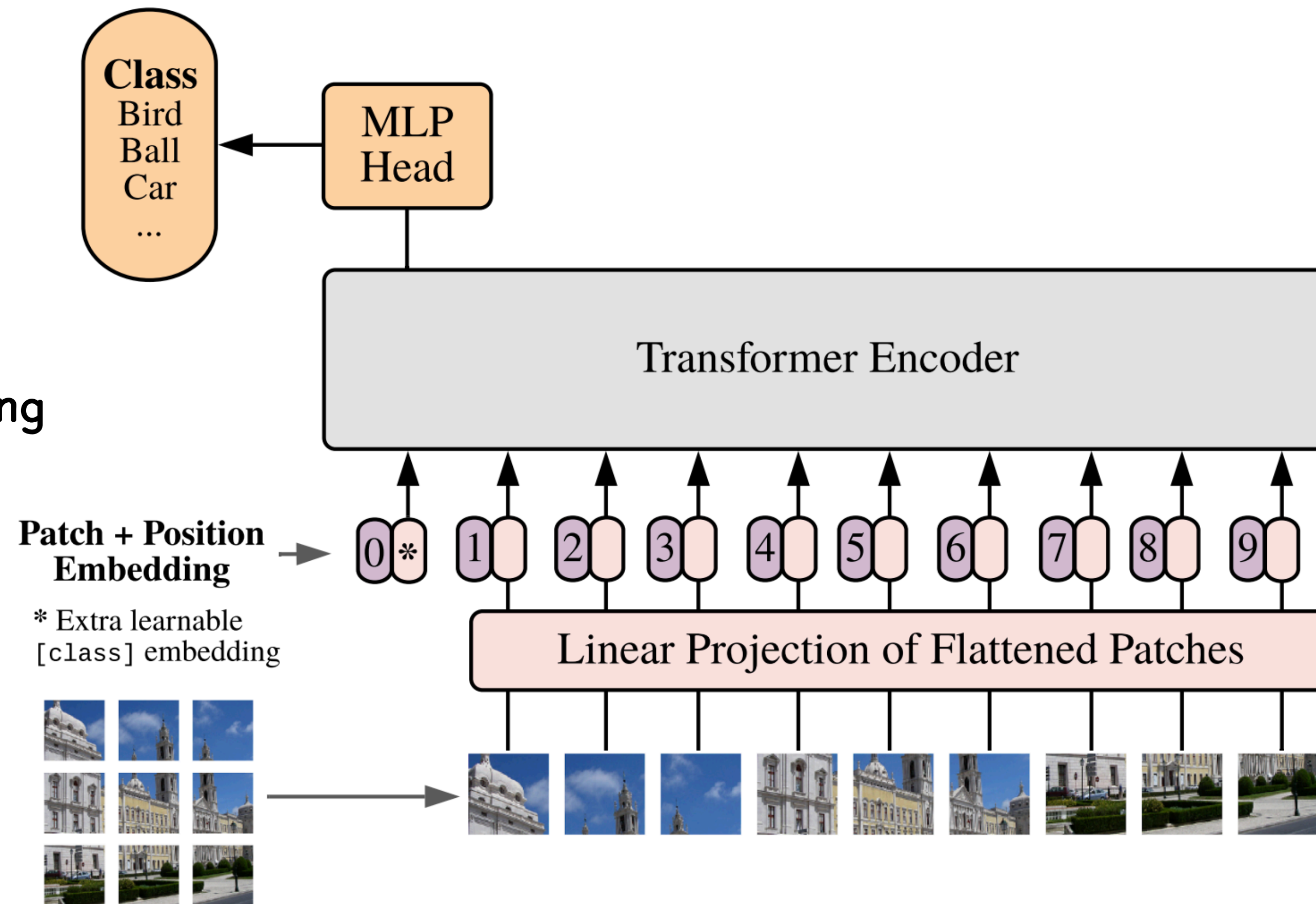


Discriminative Tasks

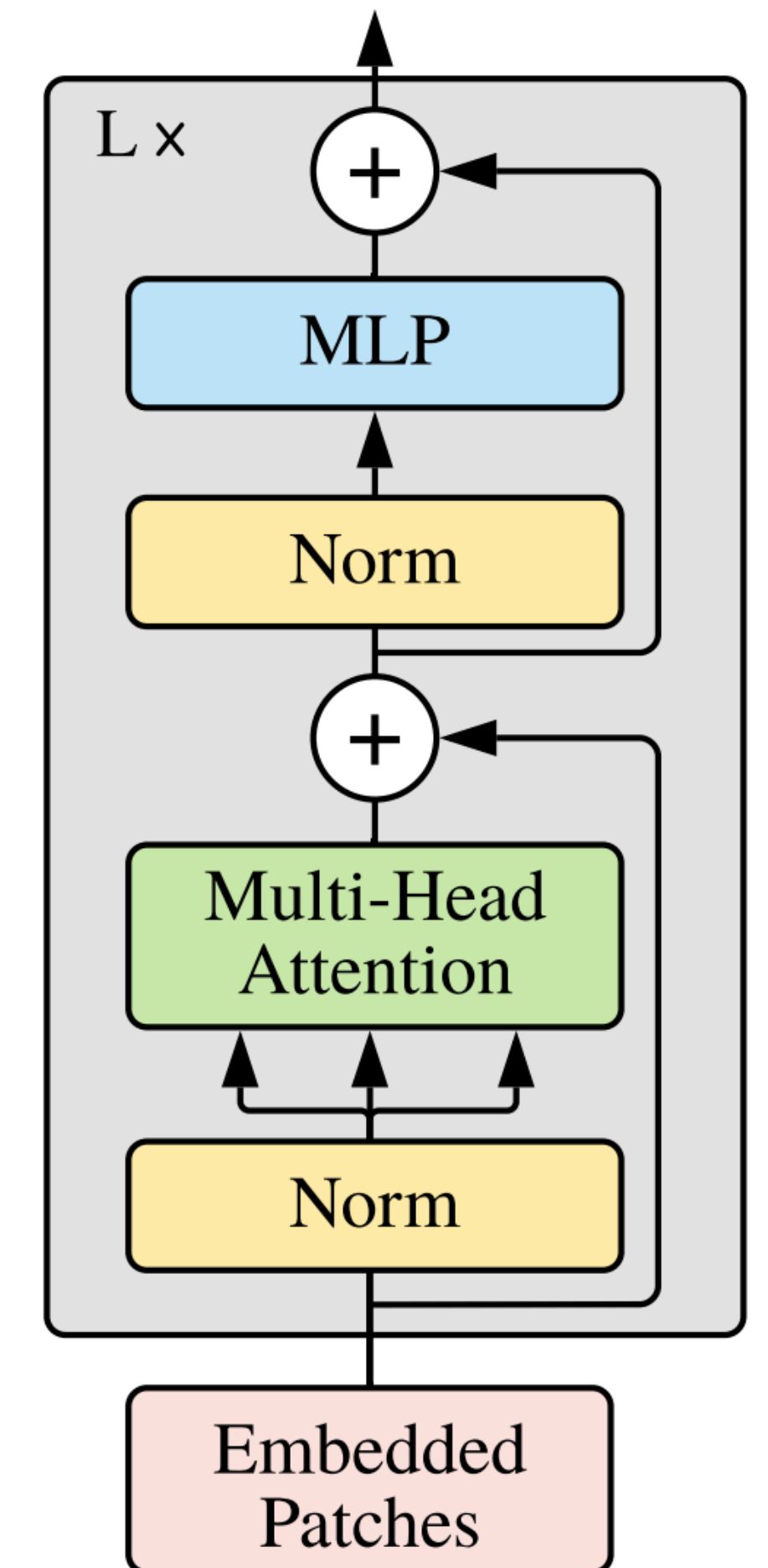
classification
regression

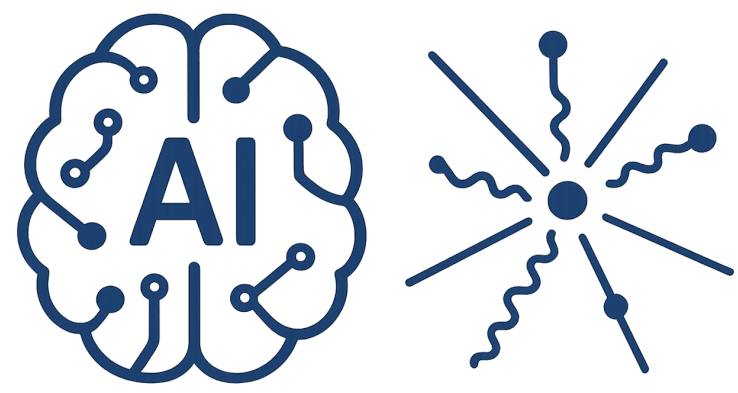
It is supervised learning

Vision Transformer (ViT)



Transformer Encoder





Discriminative Tasks

$$\mathbf{x} \in \mathbb{R}^{H \times W \times C}$$

$$\mathbf{x}_p \in \mathbb{R}^{N \times (P^2 \cdot C)}$$

D is a latent vector size

E is a trainable linear projection

$$\mathbf{z}_0 = [\mathbf{x}_{\text{class}}; \mathbf{x}_p^1 \mathbf{E}; \mathbf{x}_p^2 \mathbf{E}; \dots; \mathbf{x}_p^N \mathbf{E}] + \mathbf{E}_{\text{pos}}, \quad \mathbf{E} \in \mathbb{R}^{(P^2 \cdot C) \times D}, \mathbf{E}_{\text{pos}} \in \mathbb{R}^{(N+1) \times D}$$

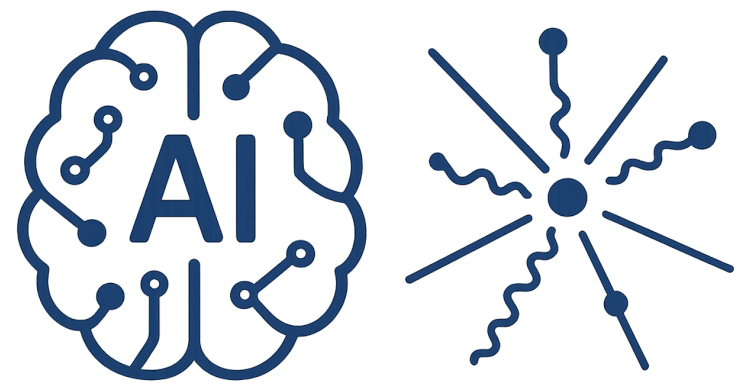
$$\mathbf{z}'_\ell = \text{MSA}(\text{LN}(\mathbf{z}_{\ell-1})) + \mathbf{z}_{\ell-1}, \quad \ell = 1 \dots L$$

$$\mathbf{z}_\ell = \text{MLP}(\text{LN}(\mathbf{z}'_\ell)) + \mathbf{z}'_\ell, \quad \ell = 1 \dots L$$

$$\mathbf{y} = \text{LN}(\mathbf{z}_L^0)$$

output of the Transformer encoder

image representation



Discriminative Tasks

$$\begin{aligned}
 \mathbf{z}_0 &= [\mathbf{x}_{\text{class}}; \mathbf{x}_p^1 \mathbf{E}; \mathbf{x}_p^2 \mathbf{E}; \cdots; \mathbf{x}_p^N \mathbf{E}] + \mathbf{E}_{pos}, & \mathbf{E} &\in \mathbb{R}^{(P^2 \cdot C) \times D}, \mathbf{E}_{pos} \in \mathbb{R}^{(N+1) \times D} \\
 \mathbf{z}'_\ell &= \text{MSA}(\text{LN}(\mathbf{z}_{\ell-1})) + \mathbf{z}_{\ell-1}, & \ell &= 1 \dots L \\
 \mathbf{z}_\ell &= \text{MLP}(\text{LN}(\mathbf{z}'_\ell)) + \mathbf{z}'_\ell, & \ell &= 1 \dots L \\
 \mathbf{y} &= \text{LN}(\mathbf{z}_L^0)
 \end{aligned}$$

$$[\mathbf{q}, \mathbf{k}, \mathbf{v}] = \mathbf{z} \mathbf{U}_{qkv}$$

$$A = \text{softmax} \left(\mathbf{q} \mathbf{k}^\top / \sqrt{D_h} \right)$$

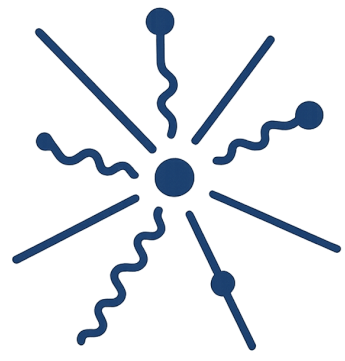
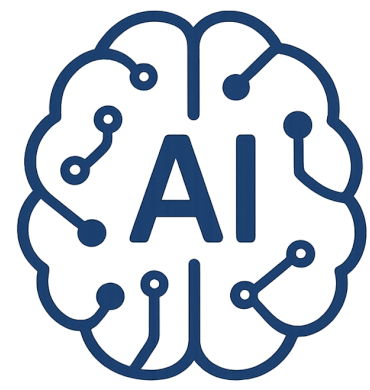
$$\text{SA}(\mathbf{z}) = A \mathbf{v}.$$

$$\mathbf{U}_{qkv} \in \mathbb{R}^{D \times 3D_h},$$

$$A \in \mathbb{R}^{(N+1) \times (N+1)},$$

$$\text{MSA}(\mathbf{z}) = [\text{SA}_1(z); \text{SA}_2(z); \cdots; \text{SA}_k(z)] \mathbf{U}_{msa}$$

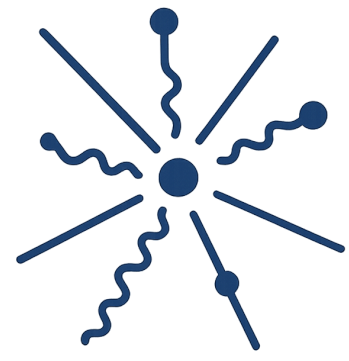
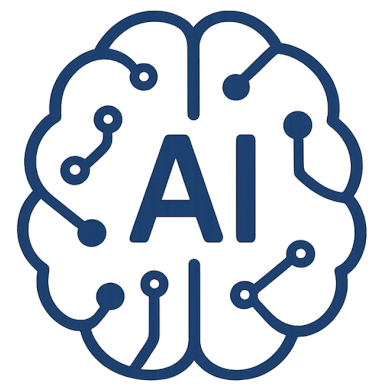
$$\mathbf{U}_{msa} \in \mathbb{R}^{k \cdot D_h \times D}$$



Discriminative Tasks

$$z_{\text{cls}}^{(L)} = f^{(L)} \left(\sum_{j=1}^N \alpha_{\text{cls},j}^{(L)} f^{(L-1)} \left(\dots \sum_{k=1}^N \alpha_{j,k}^{(1)} z_k \right) \right)$$

$$\alpha_{\text{cls},j} = \frac{\exp \left(\frac{Q_{\text{cls}} \cdot K_j}{\sqrt{d}} \right)}{\sum_{m=0}^N \exp \left(\frac{Q_{\text{cls}} \cdot K_m}{\sqrt{d}} \right)}$$



Discriminative Tasks

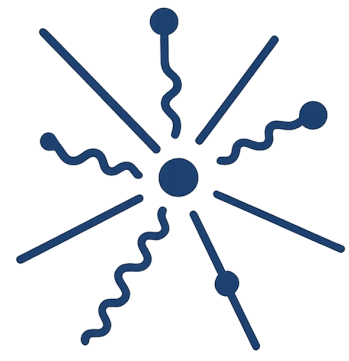
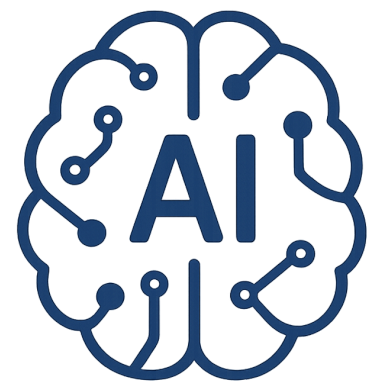
classification head

$$\text{Head}_{\text{pre}}(z_L^0) = W_2 \sigma(W_1 z_L^0 + b_1) + b_2$$

$$\text{Head}_{\text{fine}}(z_L^0) = W z_L^0 + b$$

one hidden layer at pre-training

single linear layer at fine-tuning time.



AI-Driven HEP 20

Quiz

The BERT-Large model (Devlin et al., 2018) has a maximum input length of 512 tokens, each of dimensionality $D = 1,024$ and taken from a vocabulary of 30,000. It has 24 transformer layers each with 16 self-attention heads with $D_q = D_k = D_v = 64$, and the MLP position-wise networks have two layers with 4,096 hidden nodes. Show that the total number of parameters in the BERT encoder transformer language model is approximately 340 million.